

ArbSwap

An open, oracle-anchored, two-sided market-making vault on Solana.

One hard promise: **the price you are quoted is the price you get.**

The problem

- Passive AMMs hold capital but bleed to **adverse selection** (LVR) — they are slow to reprice against informed flow.
- **PropAMMs** quote tightly and win flow, but they are closed: you cannot pool capital into them, and their fill quality is opaque.

Gap: an open vault that quotes actively — and provably bounds its own behavior.

The idea

Pooled capital + an off-chain keeper that reprices an on-chain ladder from a verified oracle.

The keeper is untrusted: the program **bounds every quote** and enforces `min_out / min_version`, so the price you are quoted is the price you get.

How it works

- **Program** (`vault/program` , Anchor): custody, Pyth-verified `update_quote` , swaps via `arb-math` , withdrawals, breakers.
- **Keeper** (`vault/keeper` , Rust): oracle tick → `arb-math` → quote payload.
- **Math** (`vault/math`): Q64.64 fixed-point ladder, shared bit-exactly by program, keeper and simulator.
- **Aggregator** (`vault/aggregator`): Jupiter-style quote engine + parity test.

Bounded and honest by construction

Every guard has a passing test and a **caught mutation** (31-guard table):

- anchor↔oracle band, level↔anchor band, min/max spread
- ladder capacity $\leq$ `utilization_max` × `available reserves`
- `update_slot` age cap; per-window one-sided flow cap
- quote expiry; `min_out` / `min_version` on every swap
- breaker is permissionless but **state-only**; `reset_breaker` admin-only

Evidence — routed world (model output)

Real price path, synthetic flow, W1–W6 (`./scripts/headline.sh`):

Venue	Volume share	Fill share	2s markout (bps)
ArbSwap	0.0%	1.6%	+1.99
B1 passive	0.3%	16.9%	−5.13
PropAMM-like	99.6%	81.5%	−1.19

The honest result

- ArbSwap **loses the routed volume share** to a tighter propAMM (~2 bps vs ~0.5–0.9 bps effective). We report this as prominently as any win.
- Its model markout is positive and better than the passive pool's — but on **synthetic flow**.
- Calibration gap (F-08): the passive benchmark is not yet the paper's benchmark.

We do not claim to beat propAMMs on routed price.

Security posture

- **No known issues in self-review, independent audit pending.**

This is a same-agent self-review, not an audit.

- Pyth-verified updates; PDA custody; every account bound to its vault.
- Keeper math parity (Rust ↔ Python) + 997 bit-exact golden vectors.
- Threat model and account-binding checklist in `docs/`.

Limits (stated up front)

- Flow is synthetic; markout/spread/PnL are **model outputs**, not measurements.
- The propAMM-like venue is a model, not a measured competitor.
- E8 (real Solana pool quote/fill gap) is **partial**: a real routed quote is recorded, the full fill gap is not measured.
- Economic value on real order flow is **not yet demonstrated**.

Roadmap

- Calibrate the passive benchmark to real adverse selection.
- Measure real-pool quote/fill gaps (E8) with execution data.
- Independent security review before any real funds.
- Jupiter listing requires a Jupiter-side Swap variant (documented).

What we claim

An **open, transparent, honest, bounded** active-liquidity vault — with a reproducible headline chart and a claims ledger where every line names its test and its limits.

Quote = fill. Bounded by construction. Honest about the rest.